XRPL Validator — Daily Engineering Report · 2026-07-21

Period: 2026-07-21 00:00 to 24:00 US Eastern (2026-07-21 04:00 to 2026-07-22 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-21): 0.99478 (93 missed of 17,817). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(node): differential_probe RPC disk cache — immutable historical lookups, DX_CACHE override** (`a585ce8`, xrpl-validator)
- **feat(xrpl-ledger): real TicketCreate + OracleSet/OracleDelete transactors; tecDST_TAG_NEEDED — corpus 97.5%** (`d28de50`, xrpl-validator) TicketCreate: batch ticket creation off tx seq (rippled CreateTicket.cpp shape), owner-dir spill via shared dir_insert, TicketCount/OwnerCount and Sequence landing past the batch. OracleSet: XLS-47 upsert with pair-level series merge and 1/2-unit reserve; new oracle keylet (0x0052 'R', mainnet-verified). Payment preclaim gains the lsfRequireDestTag check. Closes all 4 TicketCreate + 2 OracleSet SKIP-STUBs and the …
- **feat(xrpl-ledger): real NFTokenPage model — mint/burn/accept/cancel/modify on structural page keys — corpus 98.2%** (`454d6c6`, xrpl-validator) New ledger::nftpage: page key = owner(20) ++ low-96 bound, max page anchor, chain walk via PreviousPageMin, sorted insert, empty-page delete + relink. NFTokenMint composes the real NFTokenID (flags‖fee‖issuer‖taxon^scramble‖seq off FirstNFTokenSequence+MintedNFTokens); AcceptOffer takes offer INDEXES (direct + brokered), moves the token between page sets, pays XRP with the id-embedded transfer fee, and clears offer …
- **feat(xrpl-ledger): Number-exact rounding, XLS-80 domain books, multi-hop payment strands, real reserve — corpus 98.8%** (`13197e6`, xrpl-validator) st_divide/st_multiply now round half-even folding the exact division remainder (the legacy +5/+7 truncation fudges double-rounded against the new nearest pass: 0EAE58BB fixed both ways, 42071037 proved the fudge had to go). DomainID (permissioned DEX) hashes into the book base for

crossing AND placement — 105666804 replays 62/62. Payments execute multi-hop strands from the first path (chained book crossings, …

- **feat(xrpl-ledger): full AMM lifecycle parity — Create/Deposit/Withdraw on real legs — corpus 99.1%** (`a146364`, xrpl-validator) AMMCreate derives the pool account the rippled way (ripesha over sha512half(prefix0 ‖ parentHash ‖ ammKeylet), parent hash now threaded from the fixture header), creates the AMM AccountRoot + funds both legs via move_leg + mints creator LP tokens. Deposit/Withdraw move real legs between depositor and pool account and mint/burn the LP-token line (currency 0x03 ++ SHA512(minCur ‖ maxCur)[0..19], mainnet-verified). …
- **feat(xrpl-ledger): XRP autobridging for IOU↔IOU crossing; sender-reserve funding check — corpus 99.2%** (`7111b1f`, xrpl-validator) cross_bridged interleaves the direct book with the two-book XRP bridge (rippled FlowCross): peek both sources' head rates, consume one offer or bridge slice from the better one within the taker threshold; the XRP intermediate rides through the taker and nets out of the mutation set. live_head only condemns unfunded offers whose backing state is actually loaded (phantom deletions poisoned attempt 1) and never mutates …
- **fix(xrpl-ledger): me_muldiv 16-digit quotient scaling; AMM fee-adjusted spot; AMM turn in bridged crossing — corpus 99.3%** (`7d51d97`, xrpl-validator) me_muldiv truncated to ZERO when the denominator mantissa dwarfed the numerator (XRP drops x IOU 1e-11 mantissas) — the bridge slice sizing died on it; the quotient now keeps ~16 significant digits. AMM spot quality embeds the trading fee (rate = in / (out·(1−f))) and the direct-pair pool competes inside cross_bridged, anchored to the best book rate. DX_BRIDGE/DX_AMM env diagnostics stay in. 105663160 replays 65/65; …
- **feat(xrpl-ledger): rippled QualityFunction limit-trim for AMM fills — corpus 99.5%** (`96dcd62`, xrpl-validator) Ported the exact single-path offer-crossing pipeline from rippled 3.2.0 (StrandFlow limitOut + BookStep post-check): when the AMM fill would beat the taker limit, the OUT request is trimmed via the pool QualityFunction (out = (1/q − b)/m under global Upward rounding, m/b at Near), re-swapped, and the ACHIEVED quality is re-checked with a 1e-7 relative tolerance granted only to trimmed requests. Directed roundings …
- **feat(xrpl-ledger): issuer account-hops, endpoint dedup, in-flight hygiene — all corpus payments match — corpus 99.8%** (`020c568`, xrpl-validator) Path account elements through the preceding hop's ISSUER are no-op re-anchors — accepted and skipped (true third-party rippling still unsupported). Endpoint currencies named as hops collapse via chain dedup. In-flight intermediates now leave NO trace: the sender's hop lines are restored byte-exact (Sandbox::forget for never-existed lines) and a mid-chain line-creation's directory droppings (both owners' roots + last …
- **feat(xrpl-ledger): multi-path FIB AMM offers in bridged crossing — corpus stable 1107/1109** (`f2d2bcb`, xrpl-validator) IOU↔IOU crossings are always two strands on mainnet, so the direct-pair pool competes as rippled's generateFibSeqOffer: CLOB-like slices sized 0.025% of the pool's starting balances scaled by the Fibonacci sequence, judged by the slice's AVERAGE quality (spot+slippage+fee). Stops the pool from beating books it loses to by razor margins (A8830CA4 no longer touches the XAH/RLUSD pool). No corpus regressions.
- **feat(xrpl-ledger): per-leg AMM liquidity in bridged crossing — CORPUS 100% (1109/1109)** (`c333d93`, xrpl-validator) Each bridge leg is a BookStep of its own pair, so its book head competes with that pair's pool via multi-path FIB slices — mainnet filled A8830CA4's XAH leg from the XAH/XRP pool while its RLUSD leg crossed the book, and BE7884AA's missing partial fill was the same arbitration. Leg sources resolve independently (book offer or pool slice), the XRP slice flows through whichever pair wins, and pool legs advance their …

Independence program

No independence record for UTC day 2026-07-21; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Royalty Enforcement on NFT Escrow Transfers** (nft) Extend the XLS-51 NFTokenEscrow object with the same royalty accounting NFTokenAcceptOffer already performs: when an Escrow (ltESCROW) carries an sfNF
- **AMM-Triggered Stop and Take-Profit Offers** (dex) Add a tfTrigger flag to OfferCreate plus sfTriggerPrice (an STAmount ratio), sfTriggerDirection, and sfAMMID SFields on the Offer SLE, using the refer
- **Multi-Claim Pooled Escrow for Prize Pools** (gaming) Add a tfMultiClaim flag to EscrowCreate plus sfMaxClaimAmount and sfRemainingAmount SFields on the Escrow SLE. Under this flag, EscrowFinish no longer

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-21 13:01	—	xrpl-validator	`a585ce8`	feat(node): differential_probe RPC disk cache — immutable historical lookups, DX_CACHE override
07-21 19:04	—	xrpl-validator	`d28de50`	feat(xrpl-ledger): real TicketCreate + OracleSet/OracleDelete transactors; tecDST_TAG_NEEDED — corpus 97.5%

Committed	Merged into main	Repo	Commit	Subject
07-21 19:19	—	xrpl-validator	`454d6c6`	feat(xrpl-ledger): real NFTokenPage model — mint/burn/accept/cancel/modify on structural page keys — corpus 98.2%
07-21 21:30	—	xrpl-validator	`13197e6`	feat(xrpl-ledger): Number-exact rounding, XLS-80 domain books, multi-hop payment strands, real reserve — corpus 98.8%
07-21 21:35	—	xrpl-validator	`a146364`	feat(xrpl-ledger): full AMM lifecycle parity — Create/Deposit/Withdraw on real legs — corpus 99.1%
07-21 21:57	—	xrpl-validator	`7111b1f`	feat(xrpl-ledger): XRP autobridging for IOU↔IOU crossing; sender-reserve funding check — corpus 99.2%
07-21 22:28	—	xrpl-validator	`7d51d97`	fix(xrpl-ledger): me_muldiv 16-digit quotient scaling; AMM fee-adjusted spot; AMM turn in bridged crossing — corpus 99.3%
07-21 22:56	—	xrpl-validator	`96dcd62`	feat(xrpl-ledger): rippled QualityFunction limit-trim for AMM fills — corpus 99.5%
07-21 23:11	—	xrpl-validator	`020c568`	feat(xrpl-ledger): issuer account-hops, endpoint dedup, in-flight hygiene — all corpus payments match — corpus 99.8%
07-21 23:30	—	xrpl-validator	`f2d2bcb`	feat(xrpl-ledger): multi-path FIB AMM offers in bridged crossing — corpus stable 1107/1109
07-21 23:39	—	xrpl-validator	`c333d93`	feat(xrpl-ledger): per-leg AMM liquidity in bridged crossing — CORPUS 100% (1109/1109)

Times in UTC. A commit appears on the day it was committed to main or merged into it.